MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)

MG4DSEECT202 DATA SCIENCE FOR AI

Duration: 2 hours **Maximum Marks: 70**

Students should attempt at least one question from each course outcome to enhance their overall *outcome attainability*

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Part A
Multiple Choice Questions
Answer **All** Questions
Each question carries 1 mark

1	In deep learning, floating-point numbers are mainly used because they a) Reduce image size b) Allow precise mathematical computations c) Replace integer operations d) Increase memory usage	U	1
2	Identify a tensor with dimensions Height × Width × Channels is an example of a) 1D tensor b) 2D tensor c) 3D tensor d) Scalar	U	1
3	Select the main advantage of Named Tensors in PyTorch a) Faster GPU execution b) Reduced memory usage c) Improved readability and safer tensor operations d) Automatic normalization	U	1
4	Identify which tensor dimension type is commonly used to represent RGB images? a) 1D tensor b) 2D tensor c) 3D tensor d) 0D tensor	U	1
5	Show if an input image of size 32 × 32 × 3 is convolved with 16 filters of size 3 × 3, stride = 1, and zero padding = 1. What will be the output feature map size? a) 30 × 30 × 16 b) 32 × 32 × 16 c) 34 × 34 × 3 d) 32 × 32 × 3	A	2

6	Identify which convolution technique is best suited to increase the receptive field without increasing parameters? a) 1×1 convolution b) Standard convolution c) Dilated convolution d) Max pooling	A	2
7	A 1×1 convolution is mainly applied to a) Increase spatial resolution b) Reduce or expand depth (channels) c) Perform pooling d) Replace fully connected layers	A	2
8	Applying max pooling with size 2 × 2 and stride 2 on a feature map mainly results in a) Increasing depth b) Reducing spatial dimensions c) Increasing number of parameters d) Changing convolution kernel size	A	2
9	Select which key feature differentiates an RNN from a Feed Forward Neural Network? a) Use of convolution filters b) Presence of hidden layers c) Ability to retain past information using feedback connections d) Use of activation functions	K	3
10	In Back Propagation Through Time (BPTT), the network is trained by a) Updating weights randomly b) Treating each time step as an independent network c) Unfolding the RNN across time and applying backpropagation d) Freezing recurrent connections	U	3
11	Identify which LSTM component is mainly responsible for deciding what information to discard from the cell state? a) Input gate b) Forget gate c) Output gate d) Activation gate	A	3
12	Describe that auto encoders are best models to a) Perform supervised classification b) Generate random noise c) Learn compact representations by reconstructing input data d) Predict future sequences only	U	3
13	Identify the primary purpose of an attention mechanism in neural networks a) To increase the depth of the network b) to assign equal importance to all input tokens c) To focus on the most relevant parts of the input sequence d) To reduce the number of layers	U	4
14	Identify which problem in sequence models motivated the need for attention mechanisms? a) Overfitting in CNNs b) Vanishing gradients in deep CNNs c) Loss of long-range contextual information d) High memory usage in GPUs	U	4

15	Show in self-attention, the Query, Key, and Value vectors are obtained from a) Different input sequences b) The same input sequence c) Encoder and decoder separately d) Random initialization	A	4
16	Write which attention type prevents a model from accessing future tokens during training? a) Cross attention b) Global attention c) Causal attention d) Multi-head attention	A	4

1 * 16 = 16

Part B

Short Answer Questions

Answer any **6** Questions

Each question carries 5 marks

17	Describe tensors as multidimensional arrays with suitable examples.	U	1
18	Explain tensor indexing and slicing in PyTorch.	U	1
19	Map an input feature map, explain how stride and zero padding affect the output size of a convolutional layer during implementation.	A	2
20	Illustrate how max pooling and general pooling are applied in CNNs to improve translation invariance and reduce computation.	A	2
21	Explain how a Recurrent Neural Network processes sequential data and how it differs from a Feed Forward Neural Network.	U	3
22	Write the working principle of Back Propagation Through Time (BPTT) and explain why it is computationally expensive for long sequences.	An	3
23	Explain the need for attention mechanisms in sequence-to-sequence models and how they improve model performance.	U	4
24	Illustrate how self-attention works in practice and explain why multi-head attention is preferred over single-head attention.	A	4

6 * 5 = 30

Part C

Essay Questions

Answer any **2** Questions

Each question carries 12 marks

25	Explain tensor metadata (size, stride, and offset) and discuss how tensors are moved to GPU in PyTorch.	U	1
26	Illustrate how convolutional layers operate on images using 1D, 3D, and 1×1 convolutions. Describe the application of dilated convolution, pooling layers, and fully connected layers in building a ConvNet.	A	2
27	Write about the architecture and working of Long Short-Term Memory (LSTM) networks. Analyse how LSTM overcomes the limitations of basic RNNs.	An	3
28	Outline the working of the attention mechanism with a high-level overview. Analyse different types of attention mechanisms including self-attention, multi-head attention and cross attention	An	4

2 * 12 = 24